



The role of the gut microbiome in mediating neuroinflammation in immune-based neurological disorders

Claude Steriade, Nicola Segata, Deepak Saxena

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Comprehensive Epilepsy Center, Division of Epilepsy, Department of Neurology, New York University Langone Health, Grossman School of Medicine, New York University, New York, NY, USA (C Steriade MDCM); Department of Neuroscience, New York University Langone Health, Grossman School of Medicine, New York University, New York, NY, USA (C Steriade); Laboratory of Computational Metagenomics, CIBIO, University of Trento, Trento, Italy (Prof N Segata PhD); Department of Molecular Pathobiology, College of Dentistry, New York University, New York, NY, USA (Prof D Saxena PhD)

Correspondence to: Assoc Prof Claude Steriade, Comprehensive Epilepsy Center, Division of Epilepsy, Department of Neurology, New York University Langone Health, Grossman School of Medicine, New York University, New York, NY 10016, USA
claude.steriade@nyulangone.org

or Prof Deepak Saxena, Department of Molecular Pathobiology, College of Dentistry, New York University, New York, NY 10010, USA
ds100@nyu.edu

The gut microbiome can influence brain health by modulating neuroinflammation through various mechanisms, including immune regulation, the production of metabolites that affect neural function, gut and blood–brain barrier integrity, upstream effects via the vagus nerve, upstream migration of gut-resident lymphocytes to the brain, bile acid signalling, and endocrine activity. Changes in gut microbiota have been observed in demyelinating conditions, autoimmune encephalitis, and epilepsy. Gut microbiota composition changes can affect neuroinflammation, disease progression, and treatment outcomes. Advances in microbiome research have improved the potential for clinical translation of findings; but limitations persist, driven by the largely correlational nature of clinical studies and the complexity of microbiome sequencing and interpretation. At present, only the ketogenic diet is routinely recommended by clinicians, whereas other microbiome-based interventions remain investigational. Multiple strategies for manipulating the gut microbiome, including dietary changes, prebiotics, probiotics, postbiotics, and faecal microbiota transplantation, might be used as disease-modifying therapies in the future.

Introduction

Neuroinflammation is a core disease mechanism in many neurological disorders,¹ but its underlying trigger is still not completely understood, and therapeutic interventions targeting neuroinflammation have sometimes not yielded the anticipated clinical benefits in clinical trials.² The gut microbiome has gained increasing attention over the past decade, given its broad effect on multiple organs, including the brain, in part through modulation of the immune system.³ Over the past 5 years, rapid advances in microbiome sequencing methods have been translated into measures of the effects of the gut microbiome on health and inflammation, alongside growing efforts to translate these findings into clinical interventional studies. Evidence from animal and human studies suggests that gut microbiome alterations are meaningfully associated with the development and progression of neurological and psychiatric disorders, although causality remains unproven.^{3,4} Thus, the gut microbiome is a promising target for novel therapies, particularly in conditions in which neuroinflammation has a central role in disease inception and progression. However, more clinical research is needed before gut microbiome-based treatments can be used effectively in routine neurological care to modulate the inflammatory response and improve outcomes.

In this Review, we explore the relationship between the gut microbiome and neuroinflammation during disease development, focusing on conditions in which neuroinflammation is present throughout the early and late stages of disease. Specifically, we examine classic autoimmune neurological conditions—ie, multiple sclerosis and other demyelinating diseases—and autoimmune encephalitis, as well as epilepsy, since neuroinflammation contributes to epileptogenesis and seizure progression across multiple causes of seizures. First, we define key concepts pertaining to the gut microbiome and neuroinflammation, and then cover

evidence of various gut–brain axis mechanisms relevant to the neurological conditions discussed in this Review. Second, we review the experimental and human evidence of gut microbiota changes in demyelinating diseases, autoimmune encephalitis, and epilepsy, and cover potential pathways to translate these findings into treatment options. Finally, we highlight limitations and challenges in gut microbiome and neuroinflammation research, along with suggestions for future research to address existing knowledge gaps.

Conceptual definitions

Gut microbiome

The human microbiome constitutes a vast, dynamic ecosystem of trillions of microbial cells, including bacteria, archaea, viruses, and single-cell eukaryotes, which inhabit various body sites and are inextricably linked to host biology. This ecosystem is incredibly complex and, in different individuals, is highly variable in composition and abundance of single microorganisms, consisting of thousands of distinct microbial taxa. Although a substantial proportion of this microbial diversity is not taxonomically or functionally characterised, the gut microbiome is the most well studied microbiome and is increasingly well understood.⁴ Some microbiome members and ecological community configurations have been strongly associated with health outcomes, and can distinguish healthy individuals from those with medical conditions. The assembly of this personalised microbial fingerprint is initiated by maternal transmission, is maintained by microbial transmission within social interactions, and is highly modulated by diet and lifestyle.⁵ The gut microbiome can be a therapeutic target, with interventions ranging from novel live biotherapeutics to faecal microbiota transplantation.

Beyond local ecological interactions, the gut microbiome exerts systemic influence through the gut–brain axis, a bidirectional communication network linking the enteric nervous system and the CNS. This

axis integrates neural, hormonal, and immunological signals to maintain homeostasis and regulate complex behaviours. To deconstruct the complexity of the microbiome and its interactions, the field has transitioned from marker-gene surveys to high-resolution shotgun metagenomics (figure 1).^{6,7}

Neuroinflammation

In this Review, we focus on the relationship between the gut microbiome and disorders in which neuroinflammation has a central role. Neuroinflammation can happen in multiple ways. Evidence exists of blood–brain barrier (BBB) dysfunction. BBB breakdown is a hallmark of multiple sclerosis and its animal model—experimental autoimmune encephalomyelitis—and often precedes overt demyelination, indicating its role in initiating neuroinflammation.⁸ In autoimmune encephalitis, BBB disruption is a crucial mechanism enabling peripheral immune effectors and pathogenic antibodies to cross into the brain. Clinical studies showed that patients with autoimmune encephalitis and compromised BBB integrity, as measured by CSF:serum albumin ratios, had more severe clinical features and altered therapeutic responses than those with an intact BBB.⁹ Patients with

drug-resistant epilepsy also had BBB disruption on dynamic contrast-enhanced imaging,¹⁰ and elevated matrix metalloproteinases (eg, MMP9).¹¹ These clinical findings are in line with evidence from experimental models of epilepsy in BBB dysfunction, which can activate TGF- β signalling in astrocytes, alters neuronal excitability, and promotes neuroinflammation.¹²

CNS-resident immune cells—eg, microglia and astrocytes—can also become activated, typically through brain injury.¹ Traumatic brain injury can lead to neuroinflammation and subsequent epilepsy, and can also induce peripheral inflammation and changes in intestinal barrier integrity measures, indicating that the gut can mediate the relationship between peripheral and CNS inflammation.¹³ The meningeal compartment and its lymphatic system,¹⁴ the choroid plexus,¹⁵ and the glymphatic system¹⁶ have also been increasingly investigated as additional pathways for neuroinflammation.

The first two categories of conditions discussed in this Review (ie, multiple sclerosis and other demyelinating conditions as well as autoimmune encephalitis) have long been defined as neuroinflammatory.¹⁷ In addition to adaptive immunity having a central role in these

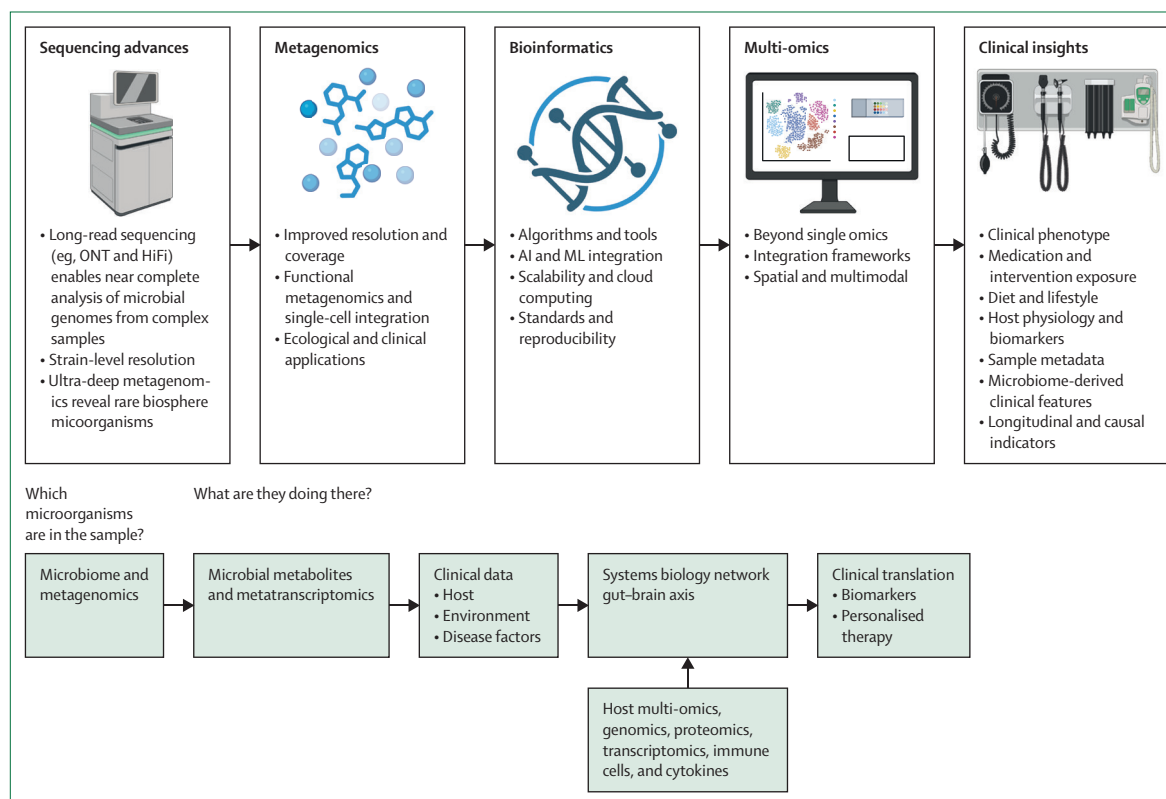


Figure 1: Advances in microbiome sequencing methodologies

High-resolution shotgun metagenomics sequences all microbial DNA in a sample and allows simultaneous profiling of what microbes are present and what functions they encode. These approaches, combined with metabolomics and other multi-omics tools, now enable the establishment of precise links between gut microbes, their metabolic products, and human disease. Bioinformatics tools can integrate metagenomics with metabolomics and metatranscriptomics to develop clinical insights and disentangle the role of host and environmental factors from that of disease in longitudinal multi-omics studies. AI=artificial intelligence. HiFi=high fidelity. ML=machine learning. ONT=Oxford Nanopore Technologies.

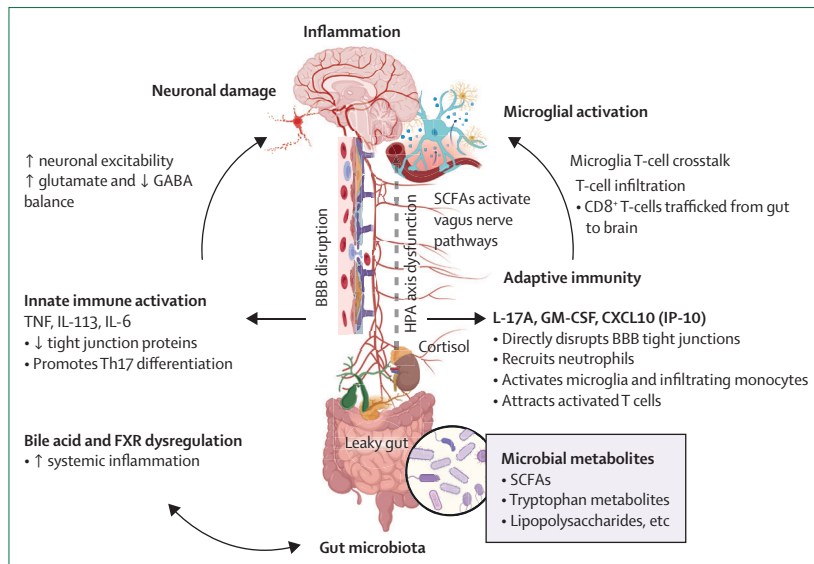


Figure 2: Mechanisms of the gut-brain axis in disease and neuroinflammation

Peripheral immune cells and signalling molecules can enter the brain through a disrupted BBB²⁵ or despite an intact BBB.²⁵ Innate and adaptive immune signalling leads to neuroinflammation, neuronal excitability, and alteration of the GABA:glutamate balance, relevant to epilepsy. Microbial pathogenicity can indirectly influence the brain via peripheral immune activation, resulting in the release of cytokines and chemokines. Gut commensal-specific CD4 T cells infiltrate the CNS regardless of antigen specificity and can be restimulated by host proteins through molecular mimicry. These T cells are primed in peripheral tissues (ie, gut and adipose tissue) by the microbiome, then trafficked to the brain. Metabolites (ie, SCFAs and tryptophan) from the gut microbiome also have an essential role in regulating neuroinflammatory processes through the gut-brain axis. A decrease in gut barrier integrity (measured by lipopolysaccharide) can increase circulating inflammatory cytokines, which can then compromise BBB integrity. The vagus nerve carries chemical signals from the gut microbiota and efferent signals, which regulate gut motility, enzyme secretion, blood flow, and immune responses. Secondary bile acid secreted by gut microbes also activates key signalling receptors, including FXR, and mitigates the immune response. The HPA axis regulates the body's response to stress and has a bidirectional relationship with the gut microbiome. BBB=blood-brain barrier. FXR=farnesoid X receptor. GABA=gamma-aminobutyric acid. HPA=hypothalamic-pituitary-adrenal axis. SCFA=short-chain fatty acid.

conditions, innate immunity is also dysregulated in both the systemic circulation and the CSF, and both histopathological analyses¹⁸ and neuroimaging methods¹⁹ have shown microglial and astrocytic activation in these conditions, albeit to different degrees according to disease type and duration. Neuroinflammation also has a key role in epileptogenesis and disease progression in epilepsy.^{20,21} Therapeutic interventions targeting inflammatory signalling attenuate seizures in experimental models of epilepsy.²² For example, peripheral inflammation in the form of bacterial endotoxin lipopolysaccharide administration induces a neuro-inflammatory response, neuronal cellular changes, and an exacerbation of epilepsy.²³ These experimental findings are mirrored by epidemiological data showing a bidirectional relationship between systemic autoimmune disorders and epilepsy.²⁴

Mechanisms of gut-brain axis in disease and neuroinflammation

In this section, we discuss mechanisms by which alterations in the gut microbiota affect neuroinflammation and brain function, relevant to the diseases highlighted in

this Review (figure 2). Beyond describing what microbes are present, modern microbiome studies increasingly measure microbial gene expression, metabolites, and host inflammatory markers in parallel. This so-called multi-omics approach allows investigators to link specific microbial pathways—such as short-chain fatty acids (SCFAs), tryptophan, or bile acid metabolism—to changes in immune profiles and clinical outcomes. Multiple lines of evidence support the mechanisms that we describe in this section, drawing on data from animal models and translational microbiome studies from human biospecimens.

Microbial pathogenicity refers to the ability of microorganisms, such as bacteria, viruses, fungi, and parasites, to cause disease in a host. Pathogenic microbes harbour factors such as toxins, surface antigens, and invasive capabilities that enable them to infect the body and trigger immune responses.²⁶ Clinical observations have shown that microbes can be directly pathogenic—eg, human herpesvirus 6 in epilepsy²⁷ and the Epstein-Barr virus in demyelinating diseases.²⁸ Growing evidence from animal models indicates that microbial pathogenicity can result in the release of cytokines and chemokines, such as interleukins (IL-1 β , IL-6) and TNF,²⁹ which then drive neuroinflammation through mechanisms previously reviewed.²⁴ Murine model evidence showed that gut commensal-specific CD4 T cells can infiltrate the CNS.^{30,31}

Metabolites and microbial products from the gut microbiome also have an essential role in regulating neuroinflammatory processes through the gut-brain axis. SCFAs, including acetate, propionate, and butyrate, are produced by bacterial fermentation of dietary fibre in the colon and have anti-inflammatory effects.³² SCFA-producing gut microbes and SCFA functional pathways have consistently been shown to be downregulated in human studies of multiple sclerosis, autoimmune encephalitis, and epilepsy (table 1, appendix pp 2–6), suggesting that this mechanism is likely directly relevant to these clinical associations. Another gut microbial metabolic pathway that modulates neuroinflammation and brain function is triggered by tryptophan, an essential dietary amino acid that is metabolised by gut microbes into compounds such as indoles, and by host pathways into kynurenine metabolites. Dysregulation of kynurenine pathways has been implicated in neuroinflammation due to an altered balance between neuroprotective metabolites (ie, kynurenine acid) and neurotoxic metabolites (ie, quinolinic acid). Quinolinic acid acts as an NMDA receptor agonist, promoting excitotoxicity, calcium influx, and neuronal hyperexcitability.³³ Elevated quinolinic acid concentrations have been detected in human multiple sclerosis lesions and CSF, where they contribute to oligodendrocyte toxicity, mitochondrial dysfunction, and oxidative stress.³⁴ Kynurenine metabolites also regulate immune responses by shaping

See Online for appendix

	Alpha diversity	Beta diversity	Gut microbial relative abundance	Association with disease state	Functional pathway findings	Comments
Demyelinating conditions						
Summary	No differences	No differences in MS vs controls, but differences between other demyelinating conditions (eg, NMOSD and ON) and controls	↓ <i>Faecalibacterium</i> (genus), <i>Faecalibacterium prausnitzii</i> (species), <i>Prevotella</i> , Lachnospiraceae (family and related groups), <i>Roseburia</i> , <i>Romboutsia</i> , and <i>Fusicatenibacter</i> ; ↑ <i>Streptococcus</i> , Verrucomicrobia, and <i>Akkermansia muciniphila</i>	↓ <i>Eubacterium</i> species associated with worse disease; effects of DMTs	Functional loss in energy, vitamin, and SCFA-associated pathways; gain in phytate degradation	Decrease in SCFA metabolites
RRMS vs healthy controls	No differences	No difference between RRMS and healthy controls, and across different types of treatment status	Genera: ↓ <i>Faecalibacterium</i> , <i>Prevotella</i> , Lachnospiraceae (family), and <i>Anaerostipes</i> ; species: ↓ <i>Faecalibacterium prausnitzii</i> and <i>Bifidobacterium adolescentis</i> , ↑ <i>Collinsella</i> , <i>Eggerthella</i> , <i>Dorea</i> sp CAG:317, and <i>Blautia</i> sp CAG:257	↓ <i>Eubacterium siraeum</i> correlated with worse EDSS	Alteration of linoleate metabolism and fatty acid metabolism	Lower concentration of SCFAs in stools of MS patients than in those of controls
RRMS, PPMS, benign MS vs healthy controls	No differences	Differences between MS subgroups; no differences compared with controls	Phyla: ↓ Bacteroidetes, Proteobacteria, ↑ Firmicutes; families: ↓ Bacteroidaceae, Tannerellaceae, Veillonellaceae, Burkholderiaceae, ↑ Ruminococcaceae, Christensenellaceae, Desulfovibrionaceae, Clostridiales, and Family XIII; species: ↓ <i>Faecalibacterium prausnitzii</i> , <i>Blautia obeum</i> , and <i>Fusicatenibacter saccharivorans</i> , ↑ <i>Akkermansia muciniphila</i> , <i>Ruthenibacterium lactatiformans</i> , and <i>Hungatella hathewayi</i>	<i>Butyricoccus</i> abundance inversely correlated with MS severity; lower disease severity with <i>Bacteroides</i> in RRMS and SCFA-producers (eg, <i>Butyrivibrio</i> , <i>Clostridium</i> , and <i>Ruminococcus</i>) in PPMS	Phytate degradation I (PWY-4702) higher in MS than in controls	..
RRMS, SPMS, atypical MS, and NMOSD vs healthy controls	No differences	Differences between each demyelinating condition and healthy controls	Genera: ↓ <i>Megamonas</i> (RRMS) and <i>Roseburia</i> (SPMS), ↑ <i>Bifidobacterium</i> (RRMS), <i>Streptococcus</i> (RRMS and SPMS), and <i>Alistipes</i> (NMOSD)	↑ <i>Streptococcus parasanguinis</i> and ↓ <i>Eubacterium hallii</i> with worse EDSS	↓ Carbohydrate metabolism in SPMS	Lower faecal SCFA concentrations in RRMS and SPMS
Paediatric MS, monophasic acute demyelinating syndrome vs healthy controls	No differences	No differences	Phyla: ↓ Actinobacteria and Firmicutes, ↑ Verrucomicrobia; genera: ↓ Lachnospiraceae (eg, <i>Anaerosporebacter</i>) and Ruminococcaceae families	Differences based on exposure to DMT	↓ Tryptophan-related NAD salvage and SCFA-producing pyruvate fermentation pathways	Independent effect of BMI on gut microbial diversity and composition
NMOSD vs healthy controls	No differences	Differences compared with controls	Genera: ↓ <i>Faecalibacterium</i> , Lachnospiraceae <i>incertae sedis</i> , <i>Prevotella</i> , <i>Blautia</i> , <i>Roseburia</i> , <i>Romboutsia</i> , <i>Coprococcus</i> , and <i>Fusicatenibacter</i> , ↑ <i>Flavonifractor</i> and <i>Streptococcus</i>	Not reported	↓ Photosynthesis, photosynthesis proteins, and thiamine metabolism	..
Demyelinating optic neuritis vs controls	Lower than controls	Differences compared with controls	Genera: ↓ <i>Bacteroides</i> , <i>Faecalibacterium</i> , <i>Roseburia</i> , <i>Parabacteroides</i> , <i>Romboutsia</i> , Lachnospiraceae ND3007 group, and <i>Alistipes</i> , ↑ <i>Blautia</i> , <i>Escherichia</i> , <i>Shigella</i> , and <i>Ruminococcus</i>	Differences after glucocorticoids and in patients who are glucocorticoid-responsive vs those resistant	Not reported	Correlation between gut microbiome and cytokines (eg, <i>Escherichia</i> and <i>Shigella</i> with MIP-3α)

(Table 1 continues on next page)

T-cell differentiation, promoting Th17 polarisation, and impairing regulatory T-cell function under inflammatory conditions.³⁵ Tryptophan pathways are disrupted across human studies of demyelinating conditions, autoimmune encephalitis, and epilepsy (table 1, appendix pp 2–6). Intestinal epithelial barrier integrity is another mechanism by which the gut microbiome modulates inflammation.³⁶

The vagus nerve is another key mediator of the effect of the gut microbiome on brain function and neuroinflammation. Vagus nerve stimulation has efficacy across multiple neurological disorders, including epilepsy, and affects neuroinflammation,³⁷ supporting the key role of this nerve in regulating the gut–brain axis. The dysregulation of enteric nervous system–neurotransmitter

signalling can disrupt epithelial and BBB integrity, promoting systemic inflammation and altering neural network activity.³⁸

Secondary bile acid secreted by gut microbes can also modulate the immune system and the gut–brain axis.³⁹ Bile acid receptor activation mitigates astrocyte and microglial pro-inflammatory polarisation in multiple sclerosis.⁴⁰ Dysbiosis-driven bile acid dysregulation has also been implicated in epilepsy-related cognitive impairment through farnesoid X receptor and ferroptosis pathways.⁴¹

The hypothalamic–pituitary–adrenal (HPA) axis regulates the body's response to stress and is modulated by the gut microbiome.⁴² In turn, stress alters the gut microbial composition and permeability. Although no

	Alpha diversity	Beta diversity	Gut microbial relative abundance	Association with disease state	Functional pathway findings	Comments
(Continued from previous page)						
Autoimmune encephalitis						
Summary across LGI1 and NMDA receptor encephalitis	Lower alpha diversity	Differences in acute cases compared with controls	Reduction in SCFA-producing genera: <i>Faecalibacterium</i> , <i>Roseburia</i> , <i>Lachnospira</i> , <i>Ruminococcus</i> , and <i>Coprococcus</i>	Not reported	Not reported	..
LGI1 antibody-mediated AE (compared with household healthy controls and age, sex, and BMI matched healthy controls)	Lower alpha diversity	Differences in immunotherapy-naïve cases compared with controls, but no differences in acute and chronic cases together compared with controls	Phyla: ↓ Bacteroidetes and Firmicutes, ↑ Proteobacteria; genera: ↓ <i>Dorea</i> , <i>Coprococcus</i> , <i>Faecalibacterium</i> , <i>Roseburia</i> , <i>Lachnospira</i> , <i>Ruminococcus</i> , and <i>Blautia</i> , ↑ <i>Enterococcus</i> and <i>Haemophilus</i> ; species: ↓ <i>Bifidobacterium longum</i> ; reduction in F:B	Not reported	↓ CoA biosynthesis and SCFA-producing microbes	No effect of ASM; effect of corticosteroids; correlation of gut microbiota with HLA-DRB1*07:01 dose
NMDA receptor encephalitis (compared with age, sex, and BMI matched healthy controls and household healthy controls)	Lower alpha diversity	Differences compared with controls	Phyla: ↓ Firmicutes, ↑ Proteobacteria; genera: ↓ <i>Faecalibacterium</i> , <i>Roseburia</i> , <i>Lachnospira</i> , <i>Ruminococcus</i> , <i>Dialister</i> , <i>Coprococcus</i> , <i>Collinsella</i> , <i>Anaerostipes</i> , and <i>Bacteroides</i> , ↑ <i>Bacteroides</i> , <i>Enterococcus</i> , <i>Escherichia</i> , <i>Veillonella</i> , <i>Streptococcus</i> , <i>Dorea</i> , <i>Scardovia</i> , <i>Clostridium</i> , and <i>Prevotella</i>	Severe disease associated with microbial dysbiosis index; <i>Granulicatella</i> predictive of good outcome	↑ Glutathione metabolism, tryptophan, and lipopolysaccharide biosynthesis proteins	..
Epilepsy						
Summary	Lower alpha diversity or no change	Differences between epilepsy and controls, but no differences between drug-responsive and drug-resistant epilepsy	Decrease in SCFA or beneficial-associated Firmicutes: <i>Faecalibacterium</i> (except ↑ in post-stroke epilepsy), <i>Ruminococcus</i> (including <i>Ruminococcus torques</i> and <i>Ruminococcus g2</i>), <i>Eubacterium</i> (<i>siraeum</i> , <i>hallii</i> , and <i>limosum</i>), <i>Lachnospira</i> , <i>Lachnospiraceae</i> , <i>Agathobacter</i> , <i>Anaerostipes</i> , <i>Phascolarctobacterium</i> , and <i>Bacteroides</i> ; increase in Proteobacteria, Actinobacteria, and other pathobionts: <i>Escherichia</i> , <i>Shigella</i> , Enterobacteriaceae-unclassified, <i>Bifidobacterium</i> , <i>Megamonas</i> , <i>Collinsella</i> , <i>Streptococcus</i> , <i>Fusobacterium</i> , <i>Sutterella</i> , <i>Halomonas</i> , <i>Megasphaera</i> , <i>Alloprevotella</i> , <i>Hungatella</i> , <i>Coriobacterium</i> , <i>Clostridium innocuum</i> , <i>Bacteroides fragilis</i> , and <i>Dialister invisus</i>	Not reported	Disruption of amino acid and glutamate-related metabolism: altered alanine, aspartate, glutamate pathways, and reduced D-glutamine or D-glutamate metabolism; reduced glutamate and quinolinic acid degradation; shifts in lipid and bile acid metabolism: reduced secondary bile acid biosynthesis and fatty acid biosynthesis	..
Epilepsy vs healthy controls	Lower or no change	Differences or no change	Phyla: ↓ Bacteroidota and Proteobacteria, ↑ Proteobacteria, Actinobacteriota, Verrucomicrobia, and Fusobacteria; genera: ↓ Lachnospiraceae-unclassified, <i>Bacteroides</i> , <i>Blautia</i> , and <i>Lachnospira</i> , ↑ <i>Blautia</i> , <i>Escherichia</i> , <i>Shigella</i> , Enterobacteriaceae-unclassified, <i>Halomonas</i> , <i>Bifidobacterium</i> , <i>Megamonas</i> , <i>Coriobacterium</i> , <i>Fusobacterium</i> , <i>Megasphaera</i> , <i>Alloprevotella</i> , and <i>Sutterella Hungatella</i>	↑ <i>Blautia</i> , <i>Bifidobacterium</i> , <i>Subdoligranulum</i> , <i>Dialister</i> , and <i>Anaerostipes</i> in drug-resistant subgroups	↓ Other glycan degradation, secondary bile acid biosynthesis, D-arginine and D-ornithine metabolism, and ↑ oxidative phosphorylation, drug metabolism, other enzymes, systemic lupus erythematosus, and beta-alanine metabolism in epilepsy vs controls	Age only confounding factor significantly correlated with increased bacterial alpha diversity in all groups
(Table 1 continues on next page)						

data exist on HPA axis–microbiome interactions in demyelinating conditions and other neuroinflammatory illnesses, preliminary data in other diseases indicate that these interactions might be of relevance. Germ-free mice show an exaggerated HPA axis response to stress, with plasma corticosterone concentrations increased approximately 2–3-times compared with conventionally-colonised controls following restraint stress. Colonisation of germ-free mice with *Bifidobacterium infantis* normalises corticosterone and adrenocorticotrophic hormone concentrations, indicating a causal role of microbiota in HPA axis programming.⁴³

In addition to directly affecting brain function and neuroinflammation, the gut microbiome can also modify the therapeutic effect of drugs. Pharmacomicrobiomics is an emerging field that studies how the gut microbiome can influence drug efficacy and toxicity, gaining traction in cancer therapies,⁴⁴ but remaining underinvestigated in neurological disorders.⁴⁵ The gut microbiome directly affects drug pharmacokinetics, including the antiseizure medication class of benzodiazepines.⁴⁶ Although the gut microbiome can modulate the variability in drug response, drugs can also modify the gut microbiome.^{47,48} Gut bacteria can chemically modify many drugs, including commonly

	Alpha diversity	Beta diversity	Gut microbial relative abundance	Association with disease state	Functional pathway findings	Comments
(Continued from previous page)						
Drug-naïve patients with epilepsy vs age matched healthy controls	Lower alpha diversity in pre-treatment epilepsy compared with healthy controls	Differences compared with controls	Phyla: Actinobacteria; genera: ↓ <i>Faecalibacterium</i> , ↑ <i>Escherichia</i> , <i>Shigella</i> , <i>Streptococcus</i> , <i>Collinsella</i> , and <i>Megamonas</i>	After initiation of oxcarbazepine, ↓ in Actinobacteria phylum as well as <i>Escherichia</i> , <i>Shigella</i> , and <i>Streptococcus</i> genera; after initiation of valproate (valproic acid), ↑ F:B; ↑ <i>Akkermansia</i> and ↓ <i>Faecalibacterium</i> after starting any ASM	Alterations in alanine, aspartate, glutamate metabolism, and citrate cycle	Specific gut microbes associated with clinical parameters post-valproate (valproic acid) initiation
Adults with drug resistance compared with drug-responsive epilepsy	No differences between drug-resistant and responsive cases	No differences between drug-resistant and responsive cases	Genera: ↓ <i>Eubacterium siraeum</i> ; species: ↓ <i>Bacteroides finegoldii</i> and <i>Ruminococcus g2</i> , ↑ <i>Negativicutes</i> (class)	Not reported	↓ D-glutamine and D-glutamate metabolism and ↑ fructose and mannose metabolism in drug-resistant epilepsy	..
Adults with temporal epilepsy, with and without anxiety	Lower alpha diversity in patients with comorbid anxiety	Differences in patients with vs without anxiety	Phyla: ↓ Firmicutes, ↑ Proteobacteria and Fusobacteriota; genera: ↓ <i>Ruminococcus torques</i> , <i>Faecalibacterium</i> , <i>Eubacterium hallii</i> , and <i>Agathobacter</i>	Not reported	Multiple pathways with differences, including biosynthesis and glycogen pathways	Study examined fungal and bacterial diversity, and found differences
Post-stroke epilepsy vs stroke and healthy controls	Higher alpha diversity in post-stroke epilepsy	Differences between post-stroke epilepsy, stroke, and healthy controls	Genera: ↓ <i>Anaerostipes</i> , ↑ <i>Clostridium innocuum</i> -group and <i>Faecalibacterium</i> in post-stroke epilepsy	Not reported	↓ fatty acid biosynthesis pathways in post-stroke epilepsy	..
Children with cerebral palsy and epilepsy vs cerebral palsy only	No differences	No differences	Species: ↓ <i>Phascolarctobacterium faecium</i> and <i>Eubacterium limosum</i> , ↑ <i>Bacteroides fragilis</i> and <i>Dialister invisus</i>	Not reported	↓ quinolinic acid degradation and glutamate degradation in patients with epilepsy	Metabolomic analyses: ↑ kynurenic acid and ↓ degradation of quinolinic acid pathway in epilepsy
NORSE vs status epilepticus of known cause vs chronic epilepsy	Higher alpha diversity in epilepsy compared with both status epilepticus cohorts	Differences between epilepsy and both status epilepticus cohorts	Species: ↑ <i>Sellimonas intestinalis</i> , ↓ <i>Nakaseomyces glabratus</i> and <i>Saccharomycodes ludwigii</i> (NORSE compared with other status epilepticus)	Prokaryotic microbial diversity associated with mortality and tube feeding	↓ valine, isoleucine, and branched-chain amino acid biosynthesis as well as pyruvate fermentation (status epilepticus), ↓ ribonucleoside use, cell wall synthesis, and chorismate biosynthesis (NORSE)	Gut microbiome composition shifted over time after status epilepticus resolution; specific cytokine-microbe associations
Alpha diversity measures the richness and evenness of microbial species within a group. Beta diversity measures how much the species composition changes between cohorts. AE=autoimmune encephalitis. ASM=antiseizure medication. CoA=coenzyme A. DMT=disease-modifying therapy. EDSS=expanded disability status scale. F:B=Firmicutes to Bacteroidetes ratio. LGI1=leucine-rich glioma inactivated 1 protein. MS=multiple sclerosis. NMOSD=neuromyelitis optica spectrum disorder. NORSE=new-onset refractory status epilepticus. ON=optic neuritis. PPMS=primary progressive multiple sclerosis. RRMS=relapsing-remitting multiple sclerosis. SCFA=short chain fatty acid. SPMS=secondary progressive multiple sclerosis.						

Table 1: Disease-specific findings in gut microbiome composition and associated functional pathways

used antiseizure medications, thereby altering their activation, inactivation, or toxicity, and ultimately affecting drug concentrations and responses.^{47,48} These effects are often species-specific—eg, *Eubacterium rectale*, a major butyrate producer, can convert some prodrugs into their active forms.^{44,49} This emerging field might help explain interindividual variability in treatment response in epilepsy and other neurological disorders.

Disease-specific gut microbiome alterations

In this section, we summarise the evidence on classic autoimmune neurological conditions (eg, demyelinating

diseases and autoimmune encephalitis), along with epilepsy—given the strong role of neuroinflammation in their inception and progression (table 1, appendix pp 2–6)—and highlight relevant correlates from experimental models.

Demyelinating conditions

A 2025 meta-analysis of studies of the gut microbiome in human cohorts with various demyelinating conditions showed differences in beta diversity (alpha diversity measures the variety of species within a single sample, community, or localised ecosystem. By contrast, beta

diversity measures the difference or change in species composition between two or more different communities or ecosystems) in multiple sclerosis and neuromyelitis optica spectrum disorders compared with healthy controls, but highlighted inconsistencies across several multiple sclerosis microbiome studies.⁵⁰ The role of non-disease covariates, including the substantial role of the recruitment site in driving variable microbiome composition in one large multisite study of multiple sclerosis,⁵¹ might account for the inconsistencies in the literature. In both demyelinating conditions, a decrease in SCFA-producing bacteria was observed,^{51,52} along with a decrease in faecal SCFA.⁵¹ These findings are mirrored by experimental evidence—susceptibility to experimental autoimmune encephalomyelitis is associated with the depletion of gut microbes implicated in SCFA metabolism.⁵³ Another mechanism of autoimmunity modulation by the gut microbiome is tryptophan metabolism, which has been shown to enhance autoimmunity by activating aryl hydrocarbon receptors and increasing T-cell production of IL-17.⁵⁴ Furthermore, tryptophan-derived metabolites are associated with the development of experimental autoimmune encephalomyelitis.⁵⁵ Gut microbial peptides have also been shown to activate myelin oligodendrocyte glycoprotein (MOG)-specific CD4⁺ T cells in vitro, which might suggest gut microbiome-induced molecular mimicry in the pathogenesis of demyelinating diseases.⁵⁶ In histopathological specimens from patients with multiple sclerosis, brain-infiltrating IgA⁺ B cells can show gut reactivity, suggesting that some clones are due to responses specific to gut bacterial surface antigens before trafficking to the brain.⁵⁷ In turn, depletion of the gut microbiome in experimental autoimmune encephalomyelitis murine models reduces pathogenic Th17 cell expression.⁵⁵

Most studies have investigated patients already on disease-modifying therapies (DMTs), but a smaller study showed that patients with multiple sclerosis who were treatment-naïve harboured a lower relative abundance of *Bacteroides fragilis* and *Bifidobacterium* spp than controls.⁵⁸ After the initiation of anti-CD20 monoclonal antibody treatment, gut microbiota composition changes became more aligned with healthy controls after starting DMT.⁵⁹ The initiation of DMTs was also followed by changes in metabolites, including SCFA changes in serum and faeces, which are specific to the type of DMT (interferon vs fingolimod).⁵¹

Autoimmune encephalitis

Surface antibody-mediated autoimmune encephalitis syndromes are prototypically cellular immunity diseases,⁶⁰ as evidenced by the efficacy of B-cell depleting therapies⁶¹ and the presence of clonally expanded B cells likely matured in the peripheral compartment in leucine-rich glioma inactivated 1 (LG1)-mediated and contactin-associated protein-like 2 (CASPR2)-mediated

encephalitis,⁶² as well as B cells initially produced peripherally in NMDA-receptor encephalitis.⁶³ Gut microbial peptide-induced immune regulation and cell trafficking to the brain, described in the previous section, might also be relevant to autoimmune encephalitis. Systemic and CSF cytokines and chemokines are also altered in various forms of autoimmune encephalitis, including LG1 antibody-mediated autoimmune encephalitis and NMDA receptor encephalitis,^{64,65} the only forms of autoimmune encephalitis in which the gut microbiome has been investigated.

At present, gut microbiome findings in autoimmune encephalitis are preliminary, given the scarcity of studies. The specific types of gut microbiota alterations differed in two studies of LG1 antibody-mediated autoimmune encephalitis, possibly due to variations in the methods used (ie, 16S rRNA gene sequencing⁶⁶ vs metagenomics⁶⁷), the nature of controls used (ie, age, sex, and BMI matched⁶⁶ vs household healthy⁶⁷), and the timing of case selection (ie, acute with pre-immunotherapy⁶⁶ vs acute with chronic⁶⁷). Functional gene prediction highlights a potential role for SCFA-producing pathways in LG1 antibody-mediated autoimmune encephalitis.⁶⁷

Patients with NMDA-receptor encephalitis not only show changes in gut microbiota composition compared with healthy controls,^{68–70} but also changes in markers of intestinal barrier integrity.⁶⁸ Specific gut microbiota correlated with cytokine markers in a cohort with NMDA receptor encephalitis of varying severity.⁶⁹ Faecal microbiota transplantation from patients with NMDA receptor encephalitis induces behavioural dysfunction in mice, albeit not a full syndrome suggestive of NMDA receptor encephalitis similar to that produced by ventricular infusions of patient-derived antibodies,⁶⁰ and increases T cells producing IL-17.⁶⁸ These findings suggest that the shifts in gut microbiota in NMDA receptor encephalitis might not simply be epiphenomena, but might be disease drivers. Notably, although each study on NMDA receptor encephalitis identified differences in gut microbiota abundance, these differences were not consistent across studies—with opposite results in the relative abundance of *Bacteroides* between two studies,^{69,70} possibly owing to small sample sizes, differences in stage of disease, effects of concomitant therapies, and the nature of controls. These caveats should be considered in future multicentre biorepository efforts for these rare disorders.⁷¹

Epilepsy

Most studies investigating the gut microbiome profile of patients with different forms of epilepsy at various ages have revealed structural differences in gut microbiome composition compared with healthy controls. However, the nature of the changes in the relative abundance of gut microbes has varied between studies (table 1, appendix pp 2–6), possibly owing to differences in age, type of epilepsy, and treatment therapy. Given that

epilepsy is a heterogeneous disease of multiple types (eg, focal, generalised) and aetiologies, microbiome-related mechanisms that vary according to epilepsy subtype are likely. Comorbidity between gastrointestinal conditions, including inflammatory bowel and coeliac diseases, and epilepsy of all causes has been shown across multiple epidemiological studies, and was previously attributed to gut dysbiosis.²⁴

Specific gut microbes (eg, *Enterobacter hormaechei* and *Candida dubliniensis*) have been associated with cytokine upregulation in status epilepticus of known cause and new-onset refractory status epilepticus,⁷² a seizure condition with high markers of neuroinflammation.⁷³ Most of the studies summarised in this Review and outlined in table 1 and the appendix have included epilepsy without a direct autoimmune cause, although neuroinflammation has been shown to be ubiquitously present across different epilepsy aetiologies that are not primarily autoimmune.⁷⁴ The occurrence of seizures themselves is an additional clinical factor that can affect the gut microbiome composition of patients with epilepsy—patients with unprovoked seizure exacerbation had a relative increase in *Fusobacteria* and *Synergistetes* in early seizure exacerbation periods compared with a later recovery time.⁷⁵ The fact that studies have not accounted for time from last seizure in their analyses might explain inconsistencies.

Three studies have specifically examined changes in the gut microbiome in patients with epilepsy who were drug-naïve after the initiation of antiseizure medications (eg, oxcarbazepine,⁷⁶ valproate [valproic acid],⁷⁷ or any antiseizure medication⁷⁸), and found alterations in gut microbiota, including an upregulation in *Akkermansia*, which is involved in gut barrier integrity and immune modulation.⁷⁹ In animal models, the ingestion of topiramate, an antiseizure medication, increases *Lactobacillus johnsonii*. In turn, co-administration of topiramate with this butyrate-producing microbe decreases seizure susceptibility to pentetrazol-induced seizures and increases cortical gamma-aminobutyric acid (GABA):glutamate ratios.⁸⁰ These findings suggest that gut microbiome manipulation could be investigated to enhance the efficacy of antiseizure medications.

The gut microbiome might also have a role in epileptogenesis. In one animal model, gut microbes associated with SCFA production differed between animals that developed epilepsy after pharmacologically-induced status epilepticus and those that did not, and were also associated with gut structural changes and increased astrogliosis and microglial activation.⁸¹ The administration of the eubiotic rifaximin in an animal model of epilepsy reduced seizure severity, improved brain and gut histopathological findings, and decreased inflammatory marker expression, supporting a causal role for gut dysbiosis in epilepsy inception and progression, and providing justification for clinical studies of microbiome-based therapeutics in epilepsy.⁸²

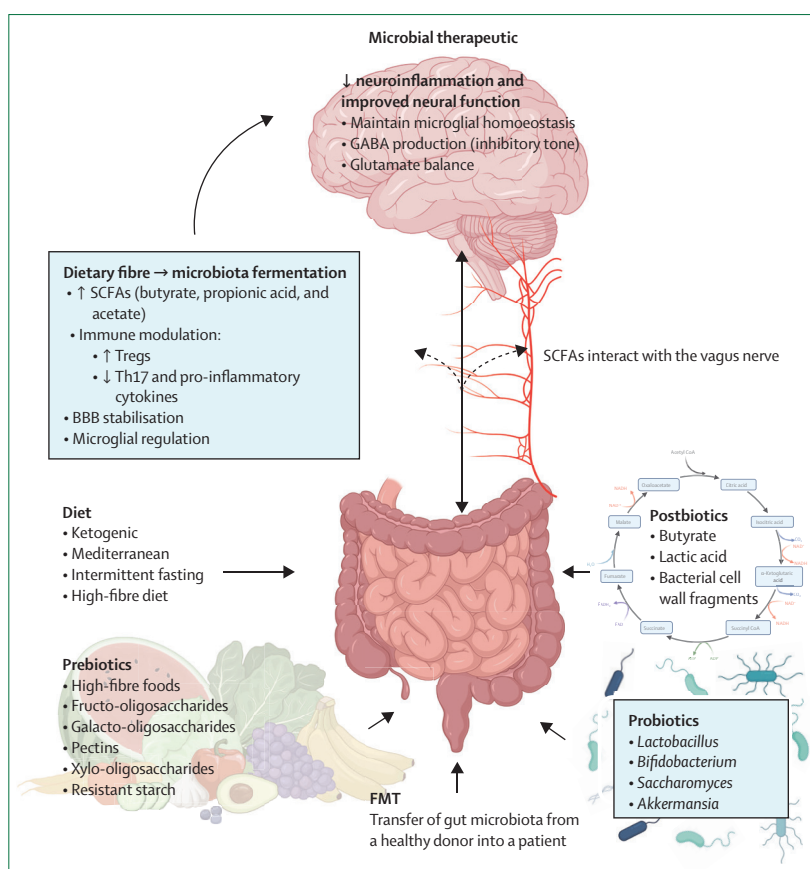


Figure 3: Approaches to microbial therapeutics

Dietary changes can modify the gut microbiome and have been increasingly studied in multiple sclerosis and epilepsy. Interventions on the gut microbiome, including prebiotics (ie, non-digestible food components that can promote specific gut microbial abundance) and probiotics (ie, specific combinations of commensal gut microbes that promote gut-brain health), have also been trialled with unknown results thus far, but many studies are active and ongoing. Postbiotics (ie, gut microbial products and metabolites) are another therapeutic approach, although few studies have investigated postbiotics. FMT, a process in which donor stool from a healthy person is administered either rectally or through an oral capsule, can modify a dysbiotic state. BBB=blood-brain barrier. FMT=faecal microbiota transplantation. GABA=gamma-aminobutyric acid. SCFA=short-chain fatty acid.

Therapeutic implications

Microbial therapeutics can influence brain function through modulation of the gut–brain axis (figure 3). In all approaches, trial design is key to address the hypothesised mechanism of action and the timecourse of effect, and measurement of biomarkers can help elucidate the cause of negative results. We summarise studies that have either been completed or have been active in the past 5 years in participants with multiple sclerosis or epilepsy (table 2, appendix pp 7–12). No such studies have been done with gut microbiome-based therapeutic approaches in other demyelinating conditions or autoimmune encephalitis. At present, the only approach regularly used in clinical care with proven efficacy is the ketogenic diet for epilepsy, but additional diet approaches, prebiotics, probiotics, postbiotics, and faecal microbiota transplantations have shown preliminary evidence in animal models or small human studies, positioning these approaches for possible advancement to larger-scale

	Participants	Intervention	Primary outcome(s)	Secondary outcome(s)	Study status and key findings
Multiple sclerosis					
Diet					
NCT03539094	RRMS	Intermittent caloric restriction vs standard diet	Change in serum leptin	Metabolic markers; immune and inflammatory profiles; body composition; exploratory cognitive outcomes; patient-reported outcomes	Completed; reduction in leptin concentrations, changes in circulating regulatory T cells, and improvement in cognitive performance
NCT04042415	RRMS; treatment-naïve	15–20% caloric restriction with or without dairy and gluten elimination vs standard diet	MS progression (NEDA)	Immune cell activity; metabolic measures; gut microbiota composition	Completed recruitment; results pending
NCT03977922	RRMS or spinal cord injury	Anti-inflammatory diet (12 weeks) vs standard diet	Composite outcome	Body composition; pain; QoL; inflammatory blood markers; lipid profile; glucose tolerance	Completed recruitment (2023); unpublished
NCT04574024	RRMS	High-fibre dietary supplement vs standard diet	Serum TNF; <i>Anaerostipes</i> abundance; faecal lipocalin-2	Not reported	Completed recruitment (2024); unpublished
NCT05712408	MS	Online-based non-restrictive diet vs standard diet	Metabolic health; bodyweight; body fat	Walking speed; cognitive assessments; QoL	Preliminary results: improved healthy eating behaviours and mental health-related QoL
NCT05007483	RRMS	Modified palaeolithic elimination diet or time-restricted olive oil-based ketogenic diet	QoL	MRI outcomes; behavioural and cognitive measures	Active; recruiting
NCT05327322	RRMS; SPMS	Low glycaemic load diet vs standard diet	Functional composite score	Cognitive and behavioural scales; sleep measures	Active; recruiting
NCT06715436	RRMS; SPMS	Ketogenic diet vs Mediterranean diet	QoL	Disability; fatigue; sleep; mood	Active; recruiting
NCT06546033	MS	Intermittent fasting vs Mediterranean diet	Blood count parameters	Sleep; QoL	Active; recruiting
NCT07033975	MS	Flavonoid-enriched Mediterranean diet vs standard diet	Gut microbiota composition	Not reported	Active; recruiting
NCT06764563	MS	Mediterranean diet vs standard diet	QoL (MSQoL-54)	Fatigue severity; disability measures; mood or depression scores; dietary adherence; anthropometric and metabolic measures	Active; recruiting
NCT06450977	MS	MIND diet vs standard diet	Cognitive performance	Diet adherence; patient-reported cognitive symptoms; mood; QoL	Active; recruiting
NCT06992115	MS	MIND diet vs standard diet	Change in neurofilament light chain	Fatigue; disability; cognitive function; blood markers of metabolism; astrocytic activation	Active; recruiting
NCT07057999	MS	MIND diet vs standard diet	MS symptom burden	QoL; mood; sleep; physical function; dietary adherence	Active; recruiting
Prebiotics and probiotics					
NCT04038541	RRMS	6 weeks Prebiotin prebiotic fibre stick pack followed by 6-week washout then probiotics (Visbiome) for 6 weeks (followed again by a 6-week washout period) vs placebo	Gut microbiome; neurofilament light chain; PBMCs	Fatigue; gastrointestinal symptoms; function	Completed recruitment (2024); pending results
NCT07304713	MS	12-week single-arm Symprove UK* vs placebo	Feasibility and acceptability	QoL; function	Active; recruiting
NCT05779449	RRMS	Prebiotic plus probiotic regimen† vs placebo	Gut microbiota; metabolites; inflammatory molecules	Neurophysiological features; physical function; disability; QoL	Active; recruiting
Antibiotics					
NCT05539729	Newly-diagnosed MS	Vancomycin 125 mg four times a day vs placebo	Butyrate-producing bacteria; butyrate concentrations; peripheral regulatory T cells	MRI measures; stool and serum SCFAs	Active; recruiting
Fecal microbiota transplantation					
NCT03975413	One case with RRMS	FMT vs placebo	Feasibility and effects on immune function	Safety; tolerability; immune biomarkers	Completed; increased abundance of commensal bacteria and SCFA metabolites, and improved gait metrics
NCT03594487	RRMS	FMT of product FMP30 (donor stool) vs placebo	Safety and feasibility; changes in gut microbiota	Immune modulation; MRI changes	Active; not recruiting
NCT04096443	MS	Oral FMT capsule vs placebo	Change in gut microbiota (16S rRNA)	QoL; disability; fatigue; immune markers	Active; not recruiting

(Table 2 continues on next page)

	Participants	Intervention	Primary outcome(s)	Secondary outcome(s)	Study status and key findings
(Continued from previous page)					
Epilepsy					
Diet					
NCT06700811	Children with genetic epilepsy with onset <6 months	Ketogenic diet (single-arm)	Achievement of ketosis and feasibility	Development of epileptic infantile spasms by 12 months; hypsarrhythmia on EEG; safety	Active; recruiting
NCT04274179	Children with classic childhood absence epilepsy	Modified Atkins vs ASM at neurologist's discretion	Change in seizure frequency	Tolerability; EEG changes	Active; recruiting
NCT06310954	Children with intractable epilepsy and ≥ 4 seizures per month	Ketogenic diet plus ASM vs standard diet plus ASM	>50% seizure reduction	Serum inflammatory markers	Active; recruiting
NCT05279118	Children with West syndrome	Ketogenic diet vs ACTH	Mean reduction in spasm rate	Spasm cessation rate; EEG; neurodevelopment	Active; not recruiting
NCT04545346	Children with ≥ 4 seizures per month	Low glutamate diet vs standard diet	Change in seizure frequency, severity, and duration	Cognition; QoL; EEG markers	Completed; no changes in seizure frequency—21% of participants responded to the diet
NCT06013761	Drug-resistant epilepsy	Intermittent fasting with and without exogenous medium-chain triglycerides vs standard diet	Change in seizure frequency	Adherence; ketosis; QoL; MRI; immune markers; cognition	Completed (2025); no published results
(Table 2 continues on next page)					

trials, and eventual implementation in clinical practice, if effective. Importantly, the long-term consequences of deliberate microbiome manipulation remain incompletely defined, with emerging concerns regarding metabolic, immunological, and potentially oncogenic complications.⁸³

Dietary interventions

The ketogenic diet is a prime example of a clinical correlate for the effect of the gut microbiome on brain health. The ketogenic diet was the first treatment for epilepsy,⁸⁴ and has remained a cornerstone of the therapeutic approach to drug resistance in multiple epilepsy types. Several studies have shown changes in the gut microbiome following initiation of the ketogenic diet,^{85–87} and the effect of this diet on seizure control has been shown to depend on an intact gut microbiome in animal models.⁸⁸ Faecal transfers from ketogenic diet-associated gut microbes result in alterations in hippocampal transcriptomics and seizure resistance.⁸⁸ A study found that serum metabolites were altered after ketogenic diet initiation in children with epilepsy, and correlations with increases in specific gut microbes, including *Faecalibacterium prausnitzii*, which is notably reduced in epilepsy, were observed (table 1, appendix pp 2–6).⁸⁵ The presence of a specific gut microbiota profile, including *Bifidobacterium*, and evidence of high

inflammation at baseline (eg, elevation of TNF in serum) have been shown to predict response to the ketogenic diet.⁸⁹ This diet has also been explored in other conditions with neuroinflammation as a means of modulating the immune response (table 2, appendix pp 7–12).⁹⁰ Despite the known effect of the ketogenic diet on the gut microbiome and inflammation, most clinical trials testing diet do not explicitly examine the effect of this diet on the gut microbiome or inflammatory markers—instead, outcome measures largely focus on symptomatic disease measures, making interpretation of negative results challenging.

Other dietary approaches have been investigated for neuroinflammatory conditions, but less evidence exists than for the ketogenic diet. Multi-omics studies have revealed associations between dietary elements, including sugar-sweetened soft drinks and meat servings, and gut microbiota profiles, metabolomics, and Th17 cells.⁹¹ A high-fibre diet is under investigation and notably, dietary fibre modifies the response to the ketogenic diet in animal models.⁹² Intermittent caloric restriction was found to improve some measures of cognitive function and increase regulatory T cells in a preliminary study of 42 people with multiple sclerosis.⁹³ The Mediterranean diet is also being assessed in several active clinical trials for multiple sclerosis (table 2, appendix pp 7–12).

	Participants	Intervention	Primary outcome(s)	Secondary outcome(s)	Study status and key findings
(Continued from previous page)					
Prebiotics and Probiotics					
NCT05160350	Drug-resistant epilepsy	Probiotic‡ vs placebo	Change in seizure frequency	Not reported	Completed recruitment (2021); no published results
NCT07100743	Drug-resistant focal epilepsy with >1 seizure per month	Probiotic§ vs placebo	QoL	Seizure frequency	Active; not recruiting
NCT05539287	Drug-resistant childhood-onset epilepsy	Probiotic¶ vs placebo	50% reduction in seizures	Changes in NLRP3 inflammatory markers	Completed (2025); unpublished
NCT05160350	Drug-resistant epilepsy	Probiotic vs placebo	Change in seizure frequency		Completed (2022); published in non-indexed source
NCT05420805	Rett syndrome	Prebiotic and postbiotic compared with postbiotic only**	Change in inflammatory markers, faecal calprotectin concentrations, and gut microbiota	Change in seizure frequency; QoL	Completed (2023); unpublished
NCT06448663	CDD	Prebiotic vs prebiotic and postbiotic††	Change in seizure frequency	Gut microbiota composition; sleep; gastrointestinal symptoms; caregiver burden	Active; recruiting
Fecal microbiota transplantation					
NCT02889627	Epilepsy with at least one seizure in the past 3 months	Repeated FMT with ~50 mL microbiota suspension infused in mid-to-lower gut	50% seizure reduction over 3 months	Gut microbiota composition	Unknown
ACTH=adrenocorticotrophic hormone. ASM=antiseizure medication. BCMC=bioscience Malaysia culture collection. CDD=cyclin-dependent kinase-like 5 deficiency disorder. EEG=electroencephalogram. FMT=faecal microbiota transplantation. HMO=human milk oligosaccharide. MCP=micro-encapsulated probiotic. MIND=Mediterranean-dietary approaches to stop hypertension intervention for neurodegenerative delay. MS=multiple sclerosis. MSQoL-54=multiple sclerosis quality of life-54. NCFM=North Carolina food microbiology. NEDA=no evidence of disease activity. PBMC=peripheral blood mononuclear cells. QoL=quality of life. RRMS=relapsing-remitting multiple sclerosis. SCFA=short chain fatty acid. SPMS=secondary progressive multiple sclerosis. *Commercially available probiotic containing <i>Lactiplantibacillus plantarum</i>, in combination with three other bacteria strains. †Regimen: days 1–15; <i>Saccharomyces boulardii</i>, <i>Bifidobacterium lactis</i> Bi-07, <i>Bifidobacterium lactis</i> BI-04, <i>Lactocaseibacillus paracasei</i> Lpc-37, <i>Lactobacillus acidophilus</i> NCFM, and prebiotics including inulin enriched with oligofructose. Days 16–365; <i>Lactobacillus acidophilus</i> NCFM, 7.5 billion <i>Bifidobacterium lactis</i> Bi-07 colony-forming units, 2.5 µg vitamin D3, 320 µg vitamin A, 100 mg threonine, 250 mg 2'-fucosyllactose (Probiactol HMO combi capsules, Metagenics). ‡Familaact 2 plus (Zist takhmir pharmaceutical company, Iran). §HEXBIO MCP granule probiotic consisting of the following probiotics: 107 mg <i>Lactobacillus acidophilus</i> (BCMC 12130), 107 mg <i>Lactobacillus casei</i> subsp (BCMC 12313), 107 mg <i>Lactobacillus lactis</i> (BCMC 12451), 107 mg <i>Bifidobacterium bifidum</i> (BCMC 02290), 107 mg <i>Bifidobacterium infantis</i> (BCMC 02129), and 107 mg <i>Bifidobacterium longum</i> (BCMC 102120). Patients are instructed to consume one sachet directly before a meal (and drink water if needed) twice daily for 2 months. ¶<i>Lactobacillus</i> species 5 billion colony-forming unit, two capsules daily. Familaact 2 plus (Zist takhmir pharmaceutical company, Iran). **Alpha-lactalbumin plus butyrate plus inulin plus fructo-oligosaccharides vs sodium butyrate plus zinc oxide. ††Alpha-lactalbumin plus inulin plus fructo-oligosaccharides vs alpha-lactalbumin plus sodium butyrate plus inulin plus fructo-oligosaccharides.					
Table 2: Clinical trials of interventions targeting the gut microbiome in participants with neurological disorders associated with neuroinflammation					

Table 2: Clinical trials of interventions targeting the gut microbiome in participants with neurological disorders associated with neuroinflammation

Prebiotics, probiotics, and postbiotics

Prebiotics, probiotics, and postbiotics have different effects in the gut microbiome (figure 3), but all can modulate the gut–brain axis by altering the gut microbiota composition and metabolic activity. Many studies administer prebiotics and postbiotics together (eg, NCT05420805 and NCT06448663 in epileptic encephalopathies), whereas others have investigated probiotics only. In human studies, the choice of prebiotics, probiotics, or postbiotics has typically not been tailored to experimental or animal evidence of gut dysbiosis specific to disease state. Notably, some completed clinical trials are unpublished (table 2, appendix pp 7–12), which might reflect a lack of efficacy in those studies and a publication bias.

In multiple sclerosis, a meta-analysis showed that probiotics alone or in combination can decrease the

severity of disease and result in immunophenotypic changes⁹⁴ and changes in gut microbiota composition, mirroring effects observed in animal models.⁹⁵ The administration of propionic acid, an SCFA, to 24 people with multiple sclerosis who were therapy-naïve, led to shifts in Treg-cell expression, mirrored by increased expression of Treg-cell inducing genes in the gut after 14 days, and improved clinical outcomes with long-term use.⁹⁶

Preclinical studies in rodent models of epilepsy show that supplementation with specific probiotics and prebiotics can reduce seizure severity, frequency, and cognitive deficits, potentially by inhibiting neuro-inflammation and oxidative stress via the gut–brain axis.⁹⁷ Experimental data suggest that probiotic supplementation, in the form of *Lachnospira eligens*, can reduce

markers of gut barrier integrity disruption and downregulate cytokine expression in epilepsy models.⁹⁸ Probiotic administration has been associated with improved gut barrier function and antiseizure effects in a meta-analysis of small-scale human studies, with some trials reporting reduced seizure frequency in drug-resistant epilepsy.⁹⁹

Faecal microbiota transplantation

Faecal microbiota transplantation affects the gut–brain axis primarily through large-scale restructuring of the intestinal microbial ecosystem. It was highly effective for recurrent *Clostridioides difficile* infection, and is now the standard of care for this condition and other gastrointestinal diseases, with established safety.¹⁰⁰ Faecal microbiota transplantation involves the transfer of complex, viable microbial communities from screened donors to recipients via lower gastrointestinal routes (eg, colonoscopy or enema), upper gastrointestinal delivery (eg, nasogastric tubes), or increasingly popular oral encapsulated formulations, each introducing distinct procedural considerations and risk profiles.¹⁰¹ Despite stringent donor screening, the risk of transmitting infectious agents, including multidrug-resistant organisms, remains a key concern, underscored by reported rare cases of bacteraemia and sepsis.¹⁰² More common adverse effects are typically mild and self-limited, including gastrointestinal discomfort, bloating, and transient alterations in bowel habits. Variability in donor selection, stool processing, and administration protocols makes reproducibility and risk stratification even more complex.¹⁰³ This evidence suggests that this intervention is not benign, but a biologically active therapy requiring standardised frameworks and longitudinal safety surveillance to fully delineate the risk–benefit profile.

Faecal microbiota transplantation can reduce gut dysbiosis-associated immune activation in animal models by normalisation of cortisol secretion and glucocorticoid receptor sensitivity, restoring intestinal epithelial barrier function and limiting the translocation of microbial products, including lipopolysaccharide, a proinflammatory molecule, into the circulation.¹⁰⁴ In experimental autoimmune encephalomyelitis models, faecal microbiota transplantation leads to changes in inflammatory and neuroprotective gene expression.¹⁰⁵ Faecal microbiota transplantation and synbiotics restore depleted SCFA-producing taxa in murine models of chronic cerebral ischaemia—an effect that might also be beneficial for the neuroinflammatory conditions discussed in this Review.¹⁰⁶ In turn, faecal microbiota transplantation was safe and tolerable in one pilot clinical study of patients with multiple sclerosis, and was associated with improved clinical outcomes.¹⁰⁷ Animal studies have also shown that faecal microbiota transplantation from donors with impending post-traumatic epilepsy modulates seizure susceptibility in murine recipients.¹⁰⁸ Given the known

Panel 1: Limitations and challenges in gut microbiome research

Causality versus correlation

- Most human studies are associative, and cannot distinguish whether microbiome changes drive neurological outcomes or result from them
- Confounding factors (eg, diet, medication, stress, and genetics) strongly influence both gut microbiota and brain function, but have infrequently been controlled for in studies

High interindividual variability

- Large person-to-person differences in microbiome composition reduce reproducibility and statistical power
- Healthy baseline microbiomes vary widely with age, geography, and lifestyle

Low translational relevance of animal models

- Rodent microbiomes differ substantially from those of humans in composition and function
- Germ-free and antibiotic-treated models might exaggerate microbiome effects and poorly reflect human physiology

Incomplete mechanistic understanding

- Multiple signalling pathways (eg, vagus nerve, immune modulation, microbial metabolites, and the hypothalamic–pituitary–adrenal axis) operate simultaneously, complicating pathway-specific conclusions
- Key microbial metabolites (eg, short-chain fatty acids, tryptophan derivatives, and bile acids) often have pleiotropic and context-dependent effects

Challenges in studying the brain correlates of microbiome compositional shifts

- Direct sampling of human brain tissue is rarely feasible, hampering mechanistic validation
- Peripheral biomarkers (eg, blood and CSF) might not accurately reflect CNS processes

Temporal and spatial dynamics

- Most studies rely on single or fixed timepoint sampling, which do not capture dynamic microbiome–brain interactions
- Regional differences within the gut (ie, small intestine vs colon) are often ignored

Technical and analytical limitations

- 16S rRNA gene sequencing has low functional resolution and metagenomics and metabolomics are costly and complex
- Batch effects, bioinformatic pipelines, and inconsistent reference databases hinder cross-study comparisons
- Low biomass samples and contaminations can introduce noise into analyses

Disease heterogeneity

- Disease subgroups might differ biologically and in microbiome signatures

Regulatory and ethical challenges

- Standardised regulatory frameworks for microbiome-based therapeutics are still evolving

effect of donor selection on efficacy in murine colitis models,¹⁰⁹ future trials on neurological conditions might have a higher likelihood of success with rational donor stratification, based on microbial and functional pathways associated with improved disease outcomes (table 1, appendix pp 2–6) and metabolite production capacity.

Conclusions and future directions

In this Review, we have shown that mounting evidence suggests an effect of the gut microbiome on

Panel 2: Future directions

Effect of clinical factors on the gut microbiome

Choice of disease subtype

Pre-defined subtypes of diseases can have different gut microbiota compositions.

Choice of control

Use of household healthy controls, the ideal group for assessing diet, lifestyle, and genetic factors independent of disease state.¹¹⁰

Tracking of concomitant therapies

Control for effect of therapeutics on the gut microbiome.

Accounting for comorbidities

Account for gastrointestinal comorbidities, including inflammatory bowel disease and coeliac disease, known to be linked to gut microbiome composition shifts.¹¹¹

Adjusting for secondary effects of disease

Shifts in gut microbiome composition can also occur due to symptoms, as observed in seizure exacerbation-related changes in the gut microbiome.⁷⁵

Effect of microbiome sequencing

Rigour and reproducibility

Validated pipelines and standardised laboratory workflows are essential to ensure that reported associations between gut microbes and neurological disease are robust.¹¹²

Accuracy

Newer long-read sequencing technologies can reconstruct microbial genomes much more completely than earlier short-read methods, and can improve strain-level and functional resolution, including more accurate detection of antimicrobial resistance genes and previously unculturable microbes.¹¹³

Effect of trial design on disease-modification mechanisms

Outcome choice

Should be adapted to the expected broader effect of gut microbiome-based therapies on brain health. For example, an epilepsy gut microbiome-based intervention would be expected to affect not only seizures, but also epilepsy comorbidities, including cognitive dysfunction and behavioural disturbances.¹¹⁴

Timing of outcome ascertainment

A clinical effect might not occur immediately after the start of therapy. A minimum run-in period before formal endpoint assessment and repeated measures of clinical and biomarker endpoints to capture delayed or progressive effects are required.

Allow for concomitant standard-of-care therapies

In conditions in which relapses are common (eg, relapses in demyelinating conditions and autoimmune encephalitis,

no improvement in treatment-refractory autoimmune encephalitis, and breakthrough seizures in epilepsy), clear guidelines on concomitant changes to standard-of-care therapies and safety exit criteria should be established. Trial designs of novel immunomodulation approaches in autoimmune encephalitis have carefully laid out such approaches to safety and concomitant treatment changes.¹¹⁵

Maximise adherence

Diet-based therapies have low long-term adherence, particularly when restrictive.¹¹⁶ Alternative dietary approaches with improved palatability, such as high-fibre, intermittent fasting, and Mediterranean diets, might be implemented more successfully. Delivery methods for faecal microbiota transplantation should also be carefully considered to improve patient acceptance.¹¹⁷

Duration of therapy and effect after withdrawal

Trials should continue to follow-up patients who withdraw from interventions to establish whether persistent benefits occur after therapy withdrawal.

Biomarkers

Ongoing trials include the measurement of changes in gut microbiota composition, along with various inflammatory biomarkers, and might help identify subgroups with a response. Some trials not showing efficacy have identified a small group of responders, in whom a subgroup analysis of biomarkers, such as baseline gut microbiota composition or inflammatory markers, might provide insights to plan for stratified, randomised clinical trials.

Focusing on metabolic effects

Because short-chain fatty acid and tryptophan pathways are relevant across multiple sclerosis, autoimmune encephalitis, and epilepsy (table 1, appendix pp 2–6), trials might prioritise interventions that affect these pathways and measure relevant metabolites longitudinally.

Faecal microbiota transplantation-specific issues

Trial design should also carefully consider the donor screening criteria according to pre-defined microbial features thought to improve disease outcomes, and standardisation of donor preparation.

Regulatory considerations

Evolving governance will need to be carefully considered for implementation.¹¹⁸

neuroinflammation. We have reviewed experimental evidence linking gut microbiota composition to the inflammatory response (ie, cytokine expression, inflammatory gene transcriptomics, and inflammatory cell composition), which is reflected by several clinical

associations (table 1, appendix pp 2–6). Although clinical data support an association between gut dysbiosis and demyelinating conditions, autoimmune encephalitis, and epilepsy, the nature of gut microbiota composition shifts has been inconsistent across studies. Multiple factors in

Search strategy and selection criteria

We searched PubMed, Google Scholar, and Web of Science for articles published in English using the terms “microbiota” OR “microbiome” OR “microbes” OR “gut”, AND “neurological” OR “epilepsy” OR “status epilepticus” OR “multiple sclerosis” OR “autoimmune encephalitis” OR “demyelinating” OR “neuroinflammation” between Jan 1, 2020, and Dec 1, 2025. The final reference list was generated on the basis of relevance to this Review, with a focus on clinical studies and animal models or mechanistic studies with direct clinical implications. The references were supplemented by the authors’ personal libraries. We also searched ClinicalTrials.gov to include active or recently completed clinical trials with gut microbiome-targeted interventions. Publications related to neurodegenerative disorders and neuroinflammation were not included, given the scope of this Review. We also included some landmark references published before 2020.

human biospecimen studies might introduce variability in findings (panel 1). Future studies should carefully consider the complex, potentially multidirectional, relationship between the gut microbiome, disease type, symptoms, and therapeutics, through longitudinal designs with repeat biospecimen sampling and appropriate measurement of relevant covariates and symptomatology to carefully disentangle the contributions of each element to the gut microbiome findings (panels 1, 2). Such longitudinal studies should integrate microbiome, metabolic, immune, and clinical phenotypic findings.

Microbiome sequencing methodologies have also varied, with studies using metagenomic shotgun sequencing and exploring multi-omics techniques to investigate mechanisms underlying gut microbiome–disease associations, shifting away from 16S rRNA gene sequencing, which might have accounted for the variability in these studies. Human DNA contamination, batch effects, and misclassification of microbial reads can introduce noise and hamper reproducibility in microbiome studies.¹¹⁹ Technical advances are likely to increase the reliability and interpretability of microbiome findings in future clinical microbiome studies, in combination with the integration of organoids for high-throughput testing of targets for intervention (appendix p 16).

Translating experimental and observational findings into human interventional studies with a high likelihood of success will require trial design that identifies a disease-targeted effect (panel 2).¹¹⁷ The past 5 years have seen a dramatic improvement in understanding the gut–brain axis in neurological disorders. With the integration of methodological advances in clinical studies, this knowledge could be translated into interventions to improve the effectiveness of available therapies in the next 5 years.

Contributors

CS and DS did the literature search, drafted the manuscript, and edited the manuscript. DS also created the original figures. NS contributed to drafting and editing of the manuscript.

Declaration of interests

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